Designing a book recommendation system in library using collaborative filtering system

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***Abstract*—The amount of data has increased, leading to the importance of data technology. The process of extracting information from a broad database is called Data Mining A recommendation system is a data filtering technique that makes suggestions to users in accordance with their requirements and areas of interest. Systems of recommendations are created to assist users in making the best product selections.It is getting harder for library patrons to select the appropriate books as the collection of books grows. A book recommendation system aids users in selecting the top book from a sizable selection. There will be many number of books on a particular subject or topic in university libraries. Students must pick the best book out of all of them. In this project, we'll create a recommendation engine based on the books that students from previous batches who received higher grades recommended. A collaborative filtering technology can be used to accomplish this. The advantage of the system is it reduces data sparsity and cold-start problems as earlier recommendation systems.**

Keywords—Librarybooks,Grades,Recommendation, Collaborative filtering.

I. INTRODUCTION

The process of extracting information from a broad database is called Data Mining. Data mining is a technique for turning unusable data into useful information. The library is a crucial tool for helping college and university students learn. Important factors of the book recommendation system are efficiency and effectiveness that could improve students' performance. The books are recommended to the students based on their grades, according to our suggested methodology. A collaborative filtering approach is used to accomplish this.

The collaborative based, content based, and hybrid techniques are the three main algorithms used in book recommendation systems. Finding similar users and recommending books to active users are the foundation of collaborative. Content-based user preference is used. Hybrid makes use of both approaches' best features. In our project, we'll use a collaborative filtering algorithm to implement a system for recommending books from the library based on student grades. In our project we are considering grades of btech(4 years) and mtech(2 years) as the datasets to recommend books for new batches

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II.LITERATURE SURVEY

A recommendation system that will recommend readers by suggesting a book which is already read by user and recommends to read a similar book can be by using deep learning at its core and is assisted by Python and Streamlit to handle the creation of front-end and to establish the CLI. This book recommender not only returns with book names and their cover page but also tries to get various information which can be the deciding factor in choosing a correct book[1].

The similarity of the content and emotional vectors found in tweets about user interests and book reviews can be used to base a book recommendation system. The cosine similarity can be determined between content vectors of the book and the interested content users to determine how similar the two pieces of content are. Each book's content vector and piece of content can be determined by finding average of all the vectors of that book[2].

The interest degree includes search times, borrowing intervals, borrowing times and renewing times according to book attributes. cosine similarity is used by collaborative filtering to calculate. Lastly, the root mean square error and the average deviation are used to calculate [3].

Popularity and inverse popularity are two factors that a personalized recommendation model looks at. The maximum number of similar documents among the available documents is determined using CF cosine similarities. This method is appropriate when there is a ton of information available and we want to extract useful information from it [4].

The duplicate of recommended items, the cold start problem data sparsity problem and low accuracy are just a few issues with the current collaborative filtering algorithm. A new algorithm for university book recommendations includes a time attenuation factor. The algorithm resolves the issue of the collaborative filtering algorithm's insufficient score[5].

An earlier study recommended a fuzzy logic-based product suggestion system for users who likes to buy books on e-commerce platforms. Using the data a decision tree model is created. The decision tree that was employed in the study was created using the rules for the fuzzy model. For consumers who want to buy a book on an e-commerce website, unsupervised learning and a fuzzy logic-based recommender system are given [6].

Hybrid approaches can be used to implement recommendation systems. Three techniques are used in hybrid approaches. The first is to create collaborative and content-based filtering separately, then combine them. The second involves combining a collaborative filtering approach with content-based capabilities and content based with collaborative based. The third method is by combining the methods into a single model. Three steps make up the user-based collaborative filtering algorithm: establish the user model, locate the set of closest neighbors, and produce recommendations. [7].

In order to make collaborative filtering more efficient Arabic sentiment analysis is applied to user reviews. Important issues with the current work were also addressed by employing efficient techniques to deal with the scalability and sparsity issues. The sentiment analysis phase and the recommendation phase make up the approach. The Arabic dataset's unique lexicon is used in the sentiment analysis phase to estimate sentiment scores. The second stage employs collaborative filtering of the item-based and singular value decomposition [8].

Machine learning modules can recommend books based on a number of features, including information from previous library loans. The method that combined the most useful features from 1.Book Titles 2. library loan records, 3.Nippon Decimal Classification categories 4.publication year and 5.borrowing frequency was found to be a). a Support Vector Machine (SVM) b.Random Forest and c.Adaboost. the simple. In order to obtain the ideal parameters, pyscript is used. C. The PPE for "SVM-AxL" was the highest, at 91.7%, 72.7%, and 74.4%, respectively. Therefore, regardless of the students' grades, "SVM-AxL" appears to be the most efficient technique[9].

The ranking of individual books for a senior comprehensive project can also serve as the basis for a recommendation system. These forecasts are based on information from the BookCrossing dataset, Google Books API, Good Reads API, and Amazon Book Reviews dataset. The raw data frames and processing should be separated using the NumPy, pandas, sklearn, and Matplotlib libraries. In order for K-means clustering to work properly in the books, the number of clusters must be determined using squared techniques[10].

Numerous techniques are enumerated, including approaches to data collection, the most regular cold start solutions, datasets that are frequently used, algorithms, and evaluation techniques. There are seven different hybridization techniques, and it has been identified that the mixed method is the most used method for gathering data. Seven different hybridization techniques are available. However, the most used hybridization techniques in recommendation systems are weighted, mixed, feature combination, and Feature Argumentation. Ask-to-rate, model-based, social network analysis-based, and solutions based on demographic data are the most frequently used approaches to solving cold start issues[11].

Two primary methods for evaluating recommendation systems are: offline evaluation and online evaluation. The simplest evaluation method is offline evaluation, which divides the data into two parts which are train and the test part. Once the training has completed,the system is tested. System interacts with actual users in online evaluation to acquire pertinent metrics from user behavior in real-time. Both offline and online evaluation have various benefits and drawbacks. First is the reproducibility factor; unlike offline evaluation, which measures user interactions in real time, online evaluation measures these interactions in real time. [12].

Item based collaborative filtering, content based collaborative filtering, custom recommender and lastly, various recommender systems are compared. A system is created such as a user can look for books of a similar genre, or based on what he has already read, he will be shown books of a similar genre from which to choose his next book. For better recommendations for user, the custom recommender will base its results on a mixture of different filtering methods[13].

A hybrid recommendation system which recommends books in advance. It is a blend of collaborative filtering and content based filtering and can be classified into three stages. In the first phase by comparing user profiles , it identifies users who are similar to the active user. It chooses the candidate's item for each similar user by getting the vectors Vc and Vm that, respectively, correlate to the user's profile and the item's contents in second phase. In the last stage, things are recommended to the target user using the Resnick prediction equation[14].

Semantic relationships are employed with content based and collaborative filtering based to get the relationships as knowledge-based book recommendation to readers in a digital library. The patterns were grouped using the Clustering method to capture how similar the books and new user were. Through a thorough tests using Information Retrieval (IR) evaluation criteria, the effectiveness of the proposed models is investigated. Recall F-Measure and Precision. Their works are put to the test in two different ways. One involves experimenting with work to compare each user to each other, and the other involves experimenting with a system to compare users' performance with clusters made up of .many. The accuracy of the book recommendation achieved is 63.84%[15].

III. System Architecture

The System Design aims to give a thorough overview of the architecture of the system, to facilitate communication and collaboration among the project team members, stakeholders, and other interested parties. This design will serve as a roadmap for the implementation and testing of the system, and will be used to guide the development of software.

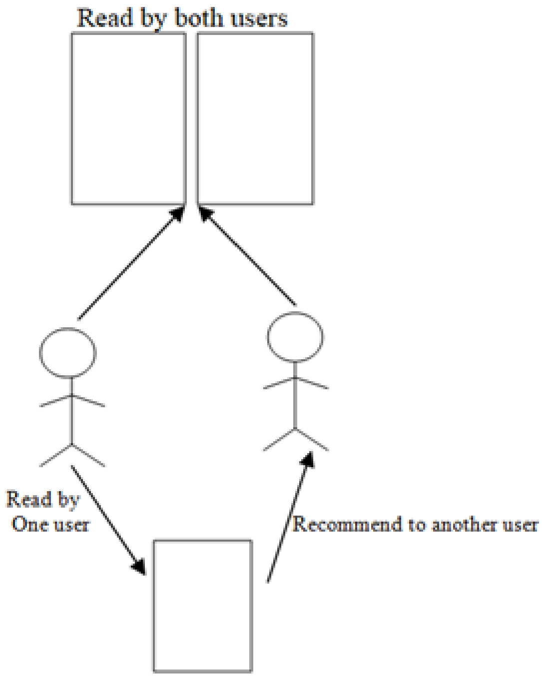

Fig 1:Basic Diagram of collaborative filtering algorithm

Collaborative filtering seeks to give users personalized recommendations based on their interests and behaviors. It is predicated on the premise that people who have similar interests and preferences today will share them tomorrow

The collaborative filtering method is used to filter the books based on the books referred by the students. Students data(grades and the books the students have recommended) is given to algorithm of collaborative filtering. Based on the results of algorithm recommendation system recommends a book to the user.

IV. IMPLEMENTATION

Flow Diagram of the system:

Figure 2 shows the flow diagram of the system. The system contains the below steps:

Step 1:Data Acquisition:

We acquired real-time data of our college, Ramaiah institute of technology through survey. The data consists of 4 years B.tech and 2 years M.tech students of batch 2019 of computer science and engineering department and the book referred by them for each subject.

3 main datasets collected are:

1.Book Dataset: Contains the title of the books of each subject referred by students

2.Grades Dataset: Contains the grades acquired by the students in each subject. Grades here are S,A,B.C,D,E and F

3.User Dataset: Contains the user details who has referred the books

Step 2:Data preprocessing:

In this step, we will check if data has any null values and eliminate them as they may affect the efficiency.

Step 3:Feature extraction:

The next step after pre-processing the gathered data is to minimize the features, also known as dimensionality reduction. High efficiency should be possible with the fewer features .In our project we can eliminate the user details which are not required for recommendation algorithm.

Step 4:Collaborative filtering algorithm:

After the final dataset is ready,collaborative filtering algorithm is applied to dataset to extract books used by the students with higher grades(in our system we have considered the grades S,A and B).

Step 5:Give Recommendations:

The system will recommend a book upon entering the semester and subject.

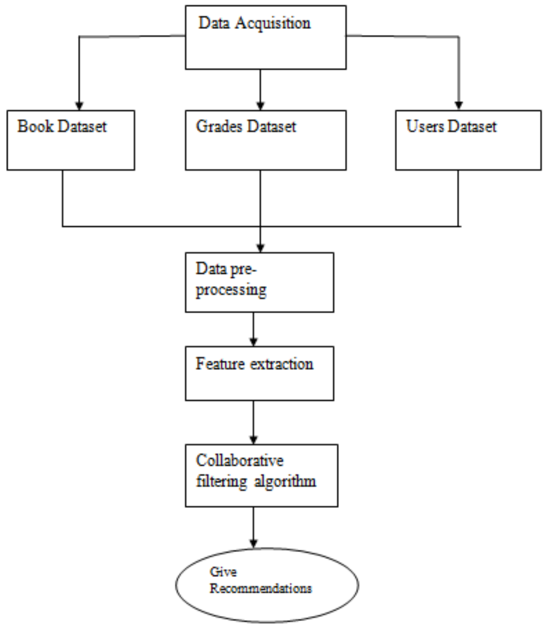

Fig 2:Flow diagram of the system

	grade	book
5	B	Rosen:Discrte Mathematics and its applications
6	A	Grimaldi:Discrete and combinatorial mathematics
7	B	Rosen:Discrte Mathematics and its applications
8	A	Grimaldi:Discrete and combinatorial mathematics
9	A	Grimaldi:Discrete and combinatorial mathematics
10	A	Grimaldi:Discrete and combinatorial mathematics
11	C	Rosen:Discrte Mathematics and its applications
12	F	Thomas koshy:Discrete Mathematics with ...
13	D	Thomas koshy:Discrete Mathematics with ...
14	A	Grimaldi:Discrete and combinatorial mathematics
15	C	Rosen:Discrte Mathematics and its applications
16	B	Rosen:Discrte Mathematics and its applications
17	S	Grimaldi:Discrete and combinatorial mathematics
18	A	Grimaldi:Discrete and combinatorial mathematics
19	A	Grimaldi:Discrete and combinatorial mathematics
20	B	Rosen:Discrte Mathematics and its applications
21	A	Grimaldi:Discrete and combinatorial mathematics
22	A	Grimaldi:Discrete and combinatorial mathematics

Fig 3:Datasets

The dataset to be processed is formed in terms of sql using sqllite as shown in figure 3.The dataset is divided into Semesters and subjects. For B.tech 7 semesters are considered and for M.tech we have further 2 sub branches i.e. Computer Science and computer networks and for each sub-branch 3 semesters are considered. Each sem of B.tech is referred as CS1,CS2 and so on and for M.tech CSE1,CSE2,CSE3 and CNE1,CNE2,CNE3 for each semester of computer Science and computer network respectively.

C. Front-end

For implementation of front end html/CSS & JavaScript has been used.

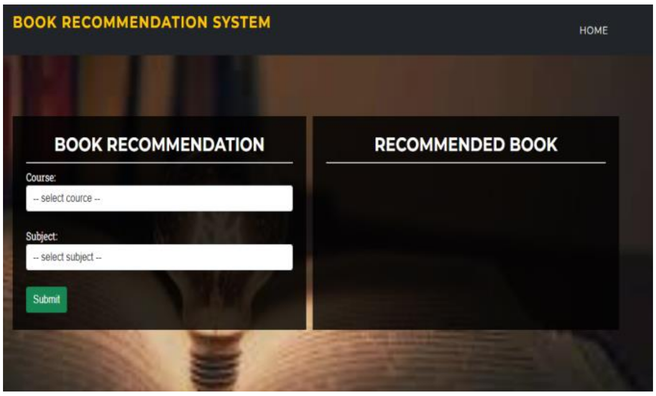

Fig 4:Front-end

Figure 4 shows the front-end of the site.

D. Back-end

For the implementation of project sqllite and python idle has been used to get the appropriate results.

The method used is simple file reading system. We have recommended the books with the students having grades S,A and B. The framework used is the flask framework which is the popular backend framework of python to create a web application.

V.RESULTS

The student needs to enter the course, semester and subject. Based on the subject chosen and the grade of students of previous batch in that particular subject the system will recommend the book to the student. Figure 5 illustrates the result when provided the semester and subject.

Fig 5:Results

VI. Conclusion

Study of literature survey helped me to able to formulate the problem statement and also understand details of different techniques used in book recommendation system. Three major algorithms used in book recommendation system are collaborative based, content based and hybrid technique. Collaborative is based on finding the similar user and recommending a book to active user. Content based used users past preference .Hybrid uses the best of both techniques. In this project we have implemented library book recommendation system using collaborative filtering algorithm with respect to grades of students. This methods will help students to score higher grades by referring best books in library .One important advantage of this system is that it does not face data sparisity and cold-start problems

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